



Markus Tatzgern

Curriculum Vitae

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Professional

- since 09/2026 **Associate Professor of Creative Technologies**, Warwick Manufacturing Group & School of Cross-Faculty Studies, University of Warwick, England
- 01/2023–08/2026 **Head of Research**, Department of Creative Technologies, Salzburg University of Applied Sciences, Austria
- 06/2018–08/2026 **FH-Professor**, Mixed Reality and Game Development, Salzburg University of Applied Sciences, Austria
- 09/2015–08/2026 **Senior Lecturer, Head of Academic Area**, Game & Mixed Reality, Salzburg University of Applied Sciences, Austria
- 09/2009–08/2015 **Research Assistant**, Christian Doppler Laboratory for Handheld Augmented Reality, Institute of Computer Graphics and Vision, Graz University of Technology, Austria

Qualifications

- 2015 Doktor der technischen Wissenschaften (**Dr.techn.**) (equivalent to Ph.D. degree), Computer Sciences, Graduated with highest distinction, Graz University of Technology, Austria, Institute of Computer Graphics and Vision, Thesis 'Situating Visualization in Augmented Reality'
- 2008 Diplom-Ingenieur (**Dipl.-Ing.**) Telematics (equivalent to Master of Science), Graduated with highest distinction, Graz University of Technology, Austria, Institute of Computer Graphics and Vision, Thesis 'A Framework for Automatically Creating 3D Explosion Diagrams'
- 2005 Bakkalaureus der techn. Wissenschaften (**Bakk.techn.**) Telematics (equivalent to Bachelor of Science), Graz University of Technology, Austria

Awards and Special Recognitions

- 2025 Honorable Mention for Best Paper at the ACM International Conference on Mobile Human-Computer Interaction (MobileHCI), Paper: You're making things AR-kward: Exploring Augmented Reality In-the-Wild

- 2024 Honorable Mention for Best Student Paper at the IEEE Symposium for Mixed and Augmented Reality (ISMAR) (**CORE A***), Paper: XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot
- 2022 Honorable Mention for Best Paper at the ACM CHI Conference on Human Factors in Computing Systems (CHI) (**CORE A***), Paper: AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality
- 2017 Honorable Mention for Best Paper at the IEEE Symposium on 3D User Interfaces (3DUI), Paper: Adaptive User Perspective Rendering for Handheld Augmented Reality
- 2025, 2026 Special Recognition for Outstanding Reviews for ACM CHI Conference on Human Factors in Computing Systems (CHI), Papers Track
- 2024 Reviewer Certificate for reviewing efforts from IEEE Transactions on Visualization and Computer Graphics (TVCG)
- 2020 Special Recognition for Outstanding Reviews for ACM Symposium on User Interface Software and Technology (UIST), Papers Track
- 2020 Special Recognition for Outstanding Reviews for IEEE Symposium for Mixed and Augmented Reality, Papers Track
- 2019, 2021, 2022 Recognition for 'highly useful' reviews for IEEE Symposium for Mixed and Augmented Reality, Journal Papers Track
- 2011, 2013, 2015 Distinct Inventor at Graz University of Technology

Publications

Journal

- 1 Gerlinde Emsenhuber, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. "See what I Mean? Mobile Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays". In: *IEEE Transactions on Visualization and Computer Graphics (TVCG), Special Issue IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* 31.11 (2025), pp. 9603–9613, **SJR Q1, CORE A***
- 2 Arno Kinzinger, Winfried J. Steiner, **Markus Tatzgern**, and Christine Vallaster. "Interactive product presentation in an immersive environment: The influence of functional control on hedonic aspects". In: *Journal of Retailing and Consumer Services* 83 (2025), p. 104156, **SJR Q1**
- 3 Arno Kinzinger, Winfried Steiner, **Markus Tatzgern**, and Christine Vallaster. "Comparing low sensory enabling (LSE) and high sensory enabling (HSE) virtual product presentation modes in e-commerce". In: *Information Systems Journal* 32.5 (2022), pp. 1034–1063, **SJR Q1**
- 4 Maria CR Harrington, **Markus Tatzgern**, Tom Langer, and John W Wenzel. "Augmented reality brings the real world into natural history dioramas with data visualizations and bioacoustics at the Carnegie Museum of Natural History". In: *Curator: The Museum Journal* 62.2 (2019), pp. 177–193, **SJR Q1**
- 5 Jacob Boesen Madsen, **Markus Tatzgern**, Claus B. Madsen, Dieter Schmalstieg, and Denis Kalkofen. "Temporal Coherence Strategies for Augmented Reality Labeling". In: *IEEE Transactions on Visualization and Computer Graphics (TVCG)* 22.4 (2016), pp. 1415–1423, **SJR Q1, CORE A***

- 6 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. "Exploring Real World Points of Interest: Design and Evaluation of Object-Centric Exploration Techniques for Augmented Reality". In: *Pervasive and Mobile Computing* 18 (2015), pp. 55–70, **SJR Q1**
- 7 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Multi-perspective compact explosion diagrams". In: *Computers & Graphics* 35.1 (2011), pp. 135–147

Conference

- 1 Irina Paraschivoiu, Thomas Layer-Wagner, Klaus Neundlinger, Simone Rack, and **Markus Tatzgern**. "Where's the Team Spirit? An Exploratory Study on Team Development Through Co-located Tablet-Based VR". in: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)*. vol. 10. 6. ACM, 2026, pp. 1–31, **CORE A**
- 2 Lucchas Ribeiro Skreinig, Denis Kalkofen, David Ritter, Peter Mohr, **Markus Tatzgern**, Dieter Schmalstieg, and Ana Stanescu. "Magnification Lenses for Small-Scale Augmented Reality Instructions". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. To appear. 2026, **CORE A***
- 3 Jan-Hendrik Sünderkamp, Helen Stefanidi, Alexander Meschtscherjakov, and **Markus Tatzgern**. "From Lab to Sidewalk: Exploring AR Interaction Techniques Outdoors and in Motion using Fitts' Law". In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025
- 4 Sebastian Hubenschmid, Johannes Zagermann, Robin Erb, Tiare Feuchtner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. "SpatialMouse : A Hybrid Pointing Device for Seamless Interaction Across 2D and 3D Spaces". In: *Proceedings of ACM Symp. on Virtual Reality Software and Technology (VRST)*. 2025
- 5 Helen Stefanidi, Jan-Hendrik Sünderkamp, **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. "You're making things AR-kward: Exploring Augmented Reality In-the-Wild". In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI)*. 2025, p. 24 **Honorable Mention for Best Paper**
- 6 Jan Hendrik Plümer, Kevin Yu, Ulrich Eck, Denis Kalkofen, Philipp Steininger, Nassir Navab, and **Markus Tatzgern**. "XR Prototyping of Mixed Reality Visualizations: Compensating Interaction Latency for a Medical Imaging Robot". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024, **CORE A*** **Honorable Mention for Best Student Paper**
- 7 Lucchas Ribeiro Skreinig, Peter Mohr, Blanca Berger, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. "Immersive Authoring by Demonstration of Industrial Procedures". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2024, pp. 1293–1302, **CORE A***
- 8 Helen Stefanidi, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Augmented Reality on the Move: A Systematic Literature Review for Vulnerable Road Users". In: *Proceedings of ACM Human-Computer Interaction (PACMHCI), Special Issue ACM International Conference on Mobile Human-Computer Interaction (MobileHCI)*. 2024
- 9 Jan Hendrik Plümer and **Markus Tatzgern**. "Towards a Framework for Validating XR Prototyping for Performance Evaluations of Simulated User Experiences". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2023, pp. 810–819, **CORE A***

- 10 Gerlinde Emsenhuber, Tobias Langlotz, Denis Kalkofen, Jonathan Sutton, and **Markus Tatzgern**. "Eye-Perspective View Management for Optical See-Through Head-Mounted Displays". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. New York, NY, USA: Association for Computing Machinery, 2023, **CORE A***
- 11 **Markus Tatzgern**, Michael Domhardt, Martin Wolf, Michael Cenger, Gerlinde Emsenhuber, Radomir Dinic, Nathalie Gerner, and Arnulf Hartl. "AirRes Mask: A Precise and Robust Virtual Reality Breathing Interface Utilizing Breathing Resistance as Output Modality". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. CHI '22. New York, NY, USA: Association for Computing Machinery, 2022, **CORE A***
Honorable Mention for Best Paper
- 12 **Markus Tatzgern** and Christoph Birgmann. "Exploring Input Approximations for Control Panels in Virtual Reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2021, **CORE A***
- 13 Masahiro Yamaguchi, Shohei Mori, Peter Mohr, **Markus Tatzgern**, Ana Stanescu, Hideo Saito, and Denis Kalkofen. "Video-Annotated Augmented Reality Assembly Tutorials". In: *Proceedings of ACM Symposium on User Interface Software and Technology (UIST)*. 2020, pp. 1010–1022, **CORE A***
- 14 Peter Mohr, **Markus Tatzgern**, Tobias Langlotz, Andreas Lang, Dieter Schmalstieg, and Denis Kalkofen. "TrackCap: Enabling Smartphones for 3D Interaction on Mobile Head-Mounted Displays". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. 2019, pp. 1–11, **CORE A***
- 15 Peter Mohr, **Markus Tatzgern**, Jens Grubert, Dieter Schmalstieg, and Denis Kalkofen. "Adaptive User Perspective Rendering for Handheld Augmented Reality". In: *IEEE Symposium on 3D User Interfaces (3DUI)*. 2017, pp. 176–181
Honorable Mention for Best Paper
- 16 Peter Mohr, David Mandl, **Markus Tatzgern**, Eduardo E. Veas, Dieter Schmalstieg, and Denis Kalkofen. "Retargeting Video Tutorials Showing Tools With Surface Contact to Augmented Reality". In: *Proceedings of ACM Conference on Human Factors in Computing Systems (CHI)*. ACM, 2017, pp. 6547–6558, **CORE A***
- 17 **Markus Tatzgern**, Valeria Orso, Denis Kalkofen, Giulio Jacucci, Luciano Gamberini, and Dieter Schmalstieg. "Adaptive information density for augmented reality displays". In: *Proceedings of IEEE Virtual Reality (VR)*. 2016, pp. 83–92, **CORE A***
- 18 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. "Hedgehog Labeling: View Management Techniques for External Labels in 3D Space". In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 1–6, **CORE A***
- 19 **Markus Tatzgern**, Raphael Grasset, Denis Kalkofen, and Dieter Schmalstieg. "Transitional Augmented Reality Navigation for Live Captured Scenes". In: *Proceedings of IEEE Virtual Reality (VR)*. 2014, pp. 21–26, **CORE A***
- 20 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Dynamic compact visualizations for augmented reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2013, **CORE A***
- 21 **Markus Tatzgern**, Raphael Grasset, Eduardo Veas, Denis Kalkofen, Hartmut Seichter, and Dieter Schmalstieg. "Exploring Distant Objects with Augmented Reality". In: *Joint Virtual Reality Conference of EGVE - EuroVR*. ed. by Betty Mohler, Bruno Raffin, Hideo Saito, and Oliver Staadt. 2013, pp. 1–11

- 22 Raphael Grasset, Tobias Langlotz, Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. "Image-Driven View Management for Augmented Reality Browsers". In: *Proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 2012, pp. 177–186, **CORE A***
- 23 **Markus Tatzgern**, Denis Kalkofen, and Dieter Schmalstieg. "Compact Explosion Diagrams". In: *Proceedings of the International Symposium on Non-Photorealistic Animation and Rendering*. 2010, pp. 17–26
- 24 Denis Kalkofen, **Markus Tatzgern**, and Dieter Schmalstieg. "Explosion Diagrams in Augmented Reality". In: *Proceedings of IEEE Virtual Reality (VR)*. 2009, pp. 71–78, **CORE A***

Posters, Workshops, Demos and Other Publications

- 1 Lucchas Ribeiro Skreinig, Ana Stanescu, Peter Mohr, Christoph Ebner, **Markus Tatzgern**, Dieter Schmalstieg, and Denis Kalkofen. "Augmented Reality Action Summaries for Situated Instructions of Repetitive Tasks". In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. Poster, to appear. 2026
- 2 Helen Stefanidi, Vincent van Rheden, Linjia He, Michelle Adiwangsa, Jan-Hendrik Sünderkamp, Alina Itzlinger, Andrii Matvienko, Julie R. Williamson, Ian Oakley, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Next Steps for Augmented Reality On-the-Move: Challenges and Opportunities". In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA)*. CHI EA '26. ACM, 2026
- 3 Sebastian Hubenschmid, Johannes Zagermann, Niklas Elmqvist, Tiare Feuchtner, Jens Grubert, **Markus Tatzgern**, Dieter Schmalstieg, and Harald Reiterer. "Revisiting Hybrid Input Devices for Immersive Analytics". In: *IEEE Conference on Human Factors in Immersive Analytics (HFIA), IEEE VIS Workshop*. 2025, pp. 29–33
- 4 Gerlinde Emsenhuber, Michael Smirnov, Tobias Langlotz, Denis Kalkofen, Stefanie Zollmann, and **Markus Tatzgern**. "Mobile Eye-perspective Rendering: Alignment of Image-based Vision Augmentations for Optical See-through Head-mounted Displays". In: *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. 2025, pp. 917–918
- 5 Helen Stefanidi, Jan-Hendrik Sünderkamp, Alina Itzlinger, **Markus Tatzgern**, and Alexander Meschtscherjakov. "Exploring AR In-the-Wild: An Autoethnographic Study at a Christmas Market". In: *Companion Publication of the 2025 ACM Designing Interactive Systems Conference (DIS)*. 2025, p. 7
- 6 Helen Stefanidi, **Markus Tatzgern**, Alina Itzlinger, and Alexander Meschtscherjakov. "Bringing Adaptive On-the-Move AR to Outdoor Environments". In: *ACM Conference on Human Factors in Computing Systems (CHI), Workshop Artificial Intelligence and Human Interfaces*. 2025
- 7 **Markus Tatzgern**, David Fasching, Vivian Gómez, Sebastián Calvache, and Pablo Figueroa. "Passive Haptic Feedback for Control Elements in Virtual Reality". In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2024, pp. 749–750
- 8 Gerlinde Emsenhuber, Michael Domhardt, Tobias Langlotz, Denis Kalkofen, and **Markus Tatzgern**. "Towards Eye-Perspective Rendering for Optical See-Through Head-Mounted Displays". In: *IEEE Virtual Reality and 3D User Interfaces - Abstracts and Workshops (VRW)*. 2022, pp. 640–641
- 9 Denis Kalkofen, Shohei Mori, and **Markus Tatzgern**. "Visualization and Graphics in Mixed Reality". In: *Eurographics - Tutorials*. The Eurographics Association, 2021

- 10 **Markus Tatzgern**, Denis Kalkofen, Raphael Grasset, and Dieter Schmalstieg. "Embedded Virtual Views for Augmented Reality Navigation". In: *Workshop on Visualization in Mixed Reality Environments, International Symposium on Mixed and Augmented Reality (ISMAR)*. Jan. 1, 2011

Patents

- 2025 Undisclosed patent pending (Qualcomm). Markus Tatzgern, Gerlinde Emsenhuber.
- 2014 Hierarchical clustering for view management augmented reality. Markus Tatzgern, Denis Kalkofen, Dieter Schmalstieg, Raphael Grasset. US Patent, Publication number 14/640,981, 2014
- 2013 Image-Driven View Management for Annotations. Raphael Grasset, Markus Tatzgern, Tobias Langlotz, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2013176830, November 2013
- 2012 Efficient Information Presentation for Augmented Reality. Markus Tatzgern, Denis Kalkofen and Dieter Schmalstieg. WO Patent, Publication number 2012033768, March 2012

Services

Chair Duties

- 2025 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Daejeon, South Korea, Science and Technology Papers Chair for Conference and Journal Track (transition to single track ISMAR)
- 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Greater Seattle Area, Science and Technology Program Chair for Conference and Journal Track
- 2024,2025 ACM Conference on Human Factors in Computing Systems (CHI), Associate Chair, Subcommittee User Experience and Usability
- 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Sydney, Poster Chair

Editorial Duties

- since 2026 Associate Editor of IEEE Transactions on Visualization and Computer Graphics
- since 2024 Editorial Board of Springer Journal Virtual Reality
- since 2021 Editorial Board of Journal Virtual Reality & Intelligent Hardware
- since 2020 Editorial Board of Augmented Reality: Review Editor for Frontiers in Virtual Reality

Committee Duties

- 2026 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2026 European Association for eXtended Reality (EuroXR) International Conference
- 2026 IEEE Visualization and Graphics Technical Community (VGTC) Virtual Reality Best Dissertation Award Committee Member
- 2023,2025,2027 IEEE Virtual Reality and 3D User Interfaces (VR), conference and journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2022 IEEE International Conference on Artificial Intelligence & Virtual Reality (AIVR)

- 2022 IEEE Virtual Reality and 3D User Interfaces (VR), journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2021 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), journal track (IEEE Transactions on Visualization and Computer Graphics)
- 2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR), conference track
- 2016, 2019-2022 IEEE Virtual Reality and 3D User Interfaces (VR)
- 2019 ACM Symposium on Virtual Reality Software and Technology (VRST)
- 2018 FH Forschungsforum, Chair: Digitized - Transformation durch Augmented und Virtual Reality

Other Roles

- 2026 AVRIL 2026, Committee Member for Competition XR Learning for German Informatics Society (GI)
- since 10/2025 Chapter Officer (Treasurer) of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter
- 2025 Founding Member of Forum New Media Austria (FNMA) Special Interest Group Immersive Education
- 2025,2026 EdTech Austria Fellowship Program - Mentor
- since 04/2025 Member of advisory board in FFG funded research project GreenTouch - XR in Ergotherapy, Austria
- since 10/2023 Scientific Collaboration and Consulting in Digital Health Technologies, Ludwig Boltzmann Institute - Digital Health and Prevention, Austria
- since 2023 Founding Member of Austrian ACM Special Interest Group on Computer-Human Interaction (SIGCHI) Chapter
- 2023 Jury Member for the Content Vienna Award funded by the Vienna Business Agency, Austria
- 2022 Expert Evaluator for Accreditation of Study Program for University of Applied Management, Germany
- 2020,2021,2026 Doctoral Consortium Mentor at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2017, 2018 Expert Evaluator for Austrian Research Promotion Agency (FFG)

Reviewing

- since 2009 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- since 2011 IEEE Virtual Reality (VR)
- since 2011 Computers & Graphics (C&G)
- since 2013 ACM Conference on Human Factors in Computing Systems (CHI)
- since 2017 IEEE Transactions on Visualization and Computer Graphics
- since 2016 ACM Symposium on Virtual Reality Software and Technology (VRST)
- 2025 Springer Journal Virtual Reality
- 2023 International Conference on Human-Computer Interaction (INTERACT)
- 2022,2023,2026 Doctoral Consortium Reviewer at IEEE International Symposium on Mixed and Augmented Reality (ISMAR)

- 2020 ACM Symposium on User Interface Software and Technology (UIST)
- 2022 IEEE Computer Graphics and Applications (CG&A)
- 2012 Eurographics
- 2012 ACM International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)
- 2015 ACM Symposium on Spatial User Interaction (SUI)
- 2015 International Conference on Mobile and Ubiquitous Multimedia (MUM)
- 2016 Journal of Virtual Reality and Broadcasting (JVRB)
- 2020 ACM Interaction Design and Children (IDC)

Teaching

Salzburg University of Applied Sciences, Bachelor Level

- 2015–2026 Game Development 1 (Lecture and Lab)
- 2016–2020 Game Development 2 (Lecture and Lab)
- 2016–2021 Computer Graphics (Lecture and Lab)
- 2016 Advanced Topics of Game Development and Augmented Reality (Lecture and Lab)
- 2016–2019 Study & Presentation Skills (Scientific Writing)
- 2020–2026 Seminar Paper Computer Science (Scientific Writing)
- 2016–2026 Bachelor Thesis Seminar
- 2024–2026 Augmented & Virtual Reality (Lecture and Lab)

Salzburg University of Applied Sciences, Master Level

- 2017–2026 Mixed Reality Technologies (Lecture and Lab)
- 2016 Mixed Reality Interaction (Lecture and Lab)
- 2017–2024 Real-Time Visualization (Lecture and Lab)
- 2020, 2025 Interaction Approaches & Technologies (Lecture and Lab)
- 2023 Applied Games (Lecture and Lab)
- 2024 Guest Lecture Game & Simulation Engineering (Lecture and Lab)
- 2019–2023 Research Seminar
- 2024–2026 Master Thesis Seminar 1
- 2024–2026 Research Method and Study Design
- 2016–2026 Master Thesis Seminar 2

Projects

In total, I have acquired more than € 3.4m for my lab since joining the Salzburg University of Applied Sciences in 2015 and have participated in project proposals that resulted in about € 8.7m of funding. In the following, the total project budget and the university share are given as 'total (share)'. Contract research for industry partners is not listed due to data protection and non-disclosure agreements.

Salzburg University of Applied Sciences

- 05/2025–04/2029 **GenAledTech - Content Creation with Generative Artificial Intelligence for Interactive Educational Technologies**
Funded by the **FFG** Austrian Research Promotion Agency / COIN Aufbau
Budget: 720.000€
Role: Principal investigator, project acquisition
- 04/2025–06/2027 **Extended Reality and Micro Learning for Vocational Training.**
Funded by the **Republic of Austria**, Federal Ministry Agriculture and Forestry, Climate and Environmental Protection, Regions and Water Management,
Partners: Adult Education Center Burgenland, University of Salzburg
Budget: 262.000€ (75.000€)
Role: Co-investigator, project acquisition
- 01/2025–12/2026 **TGuard: Combating Disinformation through the Analysis and Detection of Fake News Networks**
Funded by the **FFG** Austrian Research Promotion Agency / K-PASS 2023
Partner: Austrian Institute of Technology
Budget: 735.000€ (314.000€)
Role: Project development and acquisition as Head of Research
- 01/2024–12/2026 **FAIR AI: Fostering Austria's Innovative Strength and Research excellence in Artificial Intelligence**
Funded by the **FFG** Austrian Research Promotion Agency / Artificial Intelligence Mission Austria (AIM AT)
Lead partner: Austrian Institute of Technology
Budget: 3.0M€ (100.000€)
Role: Co-investigator, project acquisition
- 05/2022–04/2025 **CIRCUIT: Towards Comprehensive CBCT Imaging Pipelines for Real- time Acquisition, Analysis, Interaction and Visualization**
Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no FO999891105)
Industry partner: medPhoton GmbH
Budget: 470.000€
Role: Principal investigator, project acquisition
- 05/2022–04/2026 **doc.hci – Designing Meaningful Human-Technology Relations**
Funded by **FWF** Austrian Science Fund, doc.funds.connect Research partner: University of Salzburg
Budget: 1.0M€ (417.000€)
Role: Co-investigator, project acquisition
- 03/2021–02/2024 **EdTech Salzburg - Augmented Learning Lab**
Funded by Federal State Government of Salzburg, Austria
Research partner: Salzburg University of Education Stefan Zweig, University of Salzburg
Budget: 1.1M€ (393.000€)
Role: Principal investigator, project acquisition

- 06/2020–11/2023 **Adaptive Visualization for Augmented Reality**
 Funded by the Austrian Research Promotion Agency / **FFG** Bridge (no 877104), Austria
 Industry partner: Wikitude GmbH / Qualcomm Ltd.
 Budget: 383.000€
 Role: Project acquisition, principal investigator
- 03/2020–08/2022 **EdTech in the University Education of Healthcare Professionals**
 Funded by the Federal State Government of Salzburg, Austria
 Research partner: Health Sciences / Physiotherapy of Salzburg University of Applied Sciences
 Budget: 110.000€
 Role: Project acquisition, principal investigator
- 03/2020–08/2021 **Investigating Virtual Reality Environments for the Chemistry Curriculum**
 Funded by the Joachim Hertz Foundation, Germany
 Research partner: Paris Lodron University of Salzburg
 Budget: 18.000€ (11.000€)
 Role: Project acquisition, co-investigator
- 12/2017–11/2020 **Ecomedicine VR Physiology Lab: Information and communication technology drives innovation for health products and services**
 Funded by Federal State Government of Salzburg, Austria
 Research partner: Paracelsus Medical Private University
 Budget: 883.000€ (317.000€)
 Role: Project acquisition, co-investigator
- 03/2017–02/2019 **Optimizing the Online Shopping Experience using Virtual Reality**
 Funded by Federal State Government of State Salzburg, Austria
 Budget: 120.000€
 Role: Project acquisition, co-investigator
- [Graz University of Technology](#)
- 03/2010–08/2015 **Christian Doppler Laboratory for Handheld Augmented Reality**
 Funded by Christian Doppler Society
 Industry partner: Qualcomm Inc.
 Role: Early stage researcher
- 09/2009–02/2010 **IPCity**, Integrated Project on Interaction and Presence in Urban Environments
 Funded by European Union under contract FP-2004-IST-4-27571
 Role: Early stage researcher

Supervision

Official supervision and co-supervision of five PhD students, four of which are located at Salzburg University of Applied Sciences.

ongoing, PhD thesis by Gerlinde Emsenhuber,
 supervision *Eye-perspective Rendering for Optical See-through Head-mounted Displays*
 Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen

- ongoing, PhD thesis by Jan Hendrik Plümer,
supervision *Validation of XR Prototyping for Human-robot Interaction*
Located at Salzburg University of Applied Sciences, PhD programme at Graz University of Technology with Prof. Denis Kalkofen
- ongoing, PhD thesis by Jan-Hendrik Sünderkamp,
supervision *Effective XR Interaction while Being on the Move*
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov
- ongoing, PhD thesis by Faith Young,
co-supervision *Ethical AI-based Health Interventions*
Located at Salzburg University of Applied Sciences, PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov
- ongoing, PhD thesis by Eleni Stefanidi,
co-supervision *XR User Experiences in Outdoor Environments*
PhD programme at University of Salzburg with Prof. Alexander Meschtscherjakov

Regular supervision of bachelor theses (48 finished), master theses (24 finished, 4 ongoing).

--- Presentations

- 2026 Invited Talk, 'Everyday XR - Moving Out of the Lab', Wintergraph 2026, 19th Visual Computing Winter Workshop, Austria,
- 2025 Panel discussion, 'Immersion x Education for Youth', culttech summit, Vienna
- 2024 Invited Talk, 'EdTech - Augmented Learning Lab', EdTech Summit, Salzburg
- 2022 Invited talk and panel discussion, 'Trends and Potentials of EdTech', EdTech Austria Roadshow
- 2021 Invited talk, 'Supporting Primary School Education with Technology', EdTech Austria - Round Table
- 2021 Tutorial, Eurographics'21, 'Visualization and Graphics in Mixed Reality',
- 2018 Invited talk, 'Digitalization in Architecture and Construction - Use Cases of Virtual and Augmented Reality', Brennpunkt Alpines Bauen, Austria
- 2018 Invited talk, 'Virtual and Augmented Reality - Technological Maturity and Applications in Tourism', Brennpunkt eTourism, Austria
- 2018 Invited talk, 'State-of-the-Art of Virtual and Augmented Reality Technology and Applications', Austrian Armed Forces, Austria
- 2018 Invited talk, 'Virtual Reality in E-commerce', Marketing Club Salzburg, Austria
- 2018 Invited talk, 'Augmented and Virtual Reality for Medical Education and Health Care', 1. Salzburger Forum für Medizin und Digitalisierung, Paracelsus Medical University, Salzburg, Austria
- 2017 Invited talk, 'Augmented and Virtual Reality: A building block of industry 4.0', Alles Holz 4.0 oder was, Holzcluster, Salzburg, Austria
- 2017 Invited talk, 'Virtual and Augmented Reality in Education', eEducation Praxistagung, Pädagogische Hochschule Salzburg, Salzburg, Austria
- 2015 Invited talk, 'Augmented Reality Technology and Applications', Center for Human-Computer Interaction, University of Salzburg, Salzburg, Austria

- 2015 Invited talk, 'Next Generation Augmented Reality Games', Extended Games Day, Hochschule der Medien, Stuttgart, Germany

Selected Public Media

- 2025 "Artificial, intelligent and educational?" / "Künstlich, intelligent – und bildend", Salzburger Nachrichten (print, online), 16 December, Austria
- 2025 "Education in Salzburg making headway for using AI for teaching" / "Salzburger Bildung macht sich fit für KI in der Lehre", Salzburger Nachrichten (print, online), 10 November, Austria
- 2024 "FH Salzburg presents new research strategy", MyScience (online), 6 June
- 2023 "Saving time and money utilizing virtual prototyping" / "Mit virtuellen Prototypen Zeit und Kosten sparen", Salzburger Nachrichten (print), 20 October, Austria
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